Bureau
International des
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Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Helium Neon Laser (unstabilized) ()

Helium Neon Laser (unstabilized) ()



CCL-CCTF Frequency Standards Working Group

Bureau International
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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2: Helium Neon Laser (unstabilized) ()

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Appendix 2

1. CIPM recommended value [1] of the frequency

$f\left(\mathrm{HeNe}_{\mathrm{unstabilised}}\right) = 473.6127\ \mathrm{THz}$

equivalent to

$\lambda\left(\mathrm{HeNe}_{\mathrm{unstabilised}}\right) = 632.9908\ \mathrm{nm}$

with a relative standard uncertainty of 1.5×10^{-6} applies apply to the radiation in vacuum of an unstabilized helium-neon laser operating solely on the $3s_2 \rightarrow 2p_4$ transition, independent of the isotopic mixture of the neon.

This wavelength (in vacuum) value was also recommended by CIPM 2007 as a Realization of the Definition of the Metre

2. Source data

The source data are derived from an investigation carried out by the CCL and published in [2]

HeNe laser operating on the $3s_2 \rightarrow 2p_4$ transition

References

[1] Procès-Verbaux des Séances du Comité International des Poids et Mesures, 96th meeting (2007) 2009, Recommendation 2 (CI-2007): On the value and uncertainty of unstabilized He-Ne lasers, page 186. (see e.g. http://www.bipm.org/utils/en/pdf/CIPM2007-EN.pdf#page=78).

[2] J. A. Stone, J. E. Decker, P. Gill, P. Juncar, A. Lewis, G. D. Rovera, M. Viliesid: Advice from the CCL on the use of unstabilized lasers as standards of wavelength: the helium–neon laser at 633 nm. *Metrologia* **46**, 11–18 (2009).